

# HDOC Day-26

## Generation of Right-Aligned Patterns Using a Menu-Driven C Program

### Aim

To write a menu-driven C program using a switch statement and nested loops to generate different right-aligned patterns based on the user's choice. The patterns are implemented using separate parameterless functions with no return value, without using arrays.

### Algorithm

1. Start the program.
2. Display a menu containing: Right-Aligned Increasing Triangle, Right-Aligned Increasing Number Triangle, Right-Aligned Decreasing Triangle, and Exit.
3. Accept the user's menu choice.
4. Use a switch statement to process the selected option.
5. Call the corresponding parameterless function.
6. Inside the selected function, input the number of rows n.
7. Use nested loops: one to control rows, one to print spaces, and one to print stars or numbers.
8. Display the generated right-aligned pattern.
9. For an invalid menu choice, display an appropriate error message.
10. Continue displaying the menu until the user selects Exit.
11. Stop the program.

### C Program

```
#include <stdio.h>

void increasingTriangle(void);
void increasingNumberTriangle(void);
void decreasingTriangle(void);

int main()
{
    int choice;

    do
    {
        printf("\n----- RIGHT-ALIGNED PATTERNS MENU ----- \n");
        printf("1. Right-Aligned Increasing Triangle\n");
        printf("2. Right-Aligned Increasing Number Triangle\n");
        printf("3. Right-Aligned Decreasing Triangle\n");
        printf("4. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                increasingTriangle();
                break;
            case 2:
```

```

        increasingNumberTriangle();
        break;
    case 3:
        decreasingTriangle();
        break;
    case 4:
        printf("Exiting program...\n");
        break;
    default:
        printf("Invalid choice! Please enter a number from 1 to 4.\n");
    }
} while (choice != 4);

return 0;
}

void increasingTriangle(void)
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n - i; j++)
            printf(" ");
        for (j = 1; j <= i; j++)
            printf("* ");
        printf("\n");
    }
}

void increasingNumberTriangle(void)
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n - i; j++)
            printf(" ");
        for (j = 1; j <= i; j++)
            printf("%d ", j);
        printf("\n");
    }
}

void decreasingTriangle(void)
{
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = n; i >= 1; i--)
    {
        for (j = 1; j <= n - i; j++)
            printf(" ");
        for (j = 1; j <= i; j++)
            printf("* ");
        printf("\n");
    }
}

```

## Compilation/Execution Evidence

```
(kali㉿kali)-[~/Desktop/demo1]  
$ gcc HCOD26.c
```

```
(kali㉿kali)-[~/Desktop/demo1]  
$ ./a.out
```

```
——— RIGHT-ALIGNED PATTERNS MENU ———  
1. Right-Aligned Increasing Triangle  
2. Right-Aligned Increasing Number Triangle  
3. Right-Aligned Decreasing Triangle  
4. Exit  
Enter your choice: 1  
Enter number of rows: 4  
      *  
     * *  
    * * *  
   * * * *  
  
——— RIGHT-ALIGNED PATTERNS MENU ———  
1. Right-Aligned Increasing Triangle  
2. Right-Aligned Increasing Number Triangle  
3. Right-Aligned Decreasing Triangle  
4. Exit  
Enter your choice: 2  
Enter number of rows: 4  
      1  
     1 2  
    1 2 3  
   1 2 3 4  
  
——— RIGHT-ALIGNED PATTERNS MENU ———  
1. Right-Aligned Increasing Triangle  
2. Right-Aligned Increasing Number Triangle  
3. Right-Aligned Decreasing Triangle  
4. Exit  
Enter your choice: 3  
Enter number of rows: 4  
* * * *  
 * * *  
  * *  
   *
```

```

——— RIGHT-ALIGNED PATTERNS MENU ———
1. Right-Aligned Increasing Triangle
2. Right-Aligned Increasing Number Triangle
3. Right-Aligned Decreasing Triangle
4. Exit
Enter your choice: 5
Invalid choice! Please enter a number from 1 to 4.

```

```

——— RIGHT-ALIGNED PATTERNS MENU ———
1. Right-Aligned Increasing Triangle
2. Right-Aligned Increasing Number Triangle
3. Right-Aligned Decreasing Triangle
4. Exit
Enter your choice: 4
Exiting program ...

```

## Test Case Table

| Test Case | Choice | Input | Expected Output                                          |
|-----------|--------|-------|----------------------------------------------------------|
| 1         | 1      | n = 4 | Right-aligned increasing star triangle with 4 rows       |
| 2         | 2      | n = 4 | Right-aligned 1; 1 2; 1 2 3; 1 2 3 4                     |
| 3         | 3      | n = 4 | Right-aligned decreasing star triangle from 4 stars to 1 |
| 4         | 5      | -     | Invalid choice! Please enter a number from 1 to 4.       |
| 5         | 4      | -     | Program exits                                            |

## Result

The menu-driven C program was successfully implemented using switch-case statements, parameterless functions, and nested loops without arrays. It generates the required right-aligned increasing star triangle, right-aligned increasing number triangle, and right-aligned decreasing star triangle. The program also handles invalid menu choices and continues execution until the user selects the Exit option.